

Ezra Silver-Isenstadt

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EDUCATION

The University of Maryland, College Park | College Park, MD

B.S. in Computer Science, B.M. in Music Composition, Minor in Philosophy | GPA: 3.87/4.0

May 2026

Banneker/Key Scholar: Awarded all-expenses merit scholarship to UMD; featured as model B/K student for new recipients.

Relevant Courses: Object-Oriented Programming I & II (Java), Introduction to Computer Systems (C),
Organization of Programming Languages (OCaml, Rust), Discrete Structures, Algorithms,
Web App Development (HTML/CSS/JS, React, Node.js, MongoDB), Advanced Data Structures (Python)

TECHNICAL SKILLS

Languages: Java (Proficient), C (Proficient), Python*, HTML/CSS/JS*, Rust (Intermediate), OCaml (Intermediate)

Tools/Frameworks: Git, Unix Command Line, Neovim, MongoDB*, React*, Node.js*, VSCode, Eclipse

* = Proficient by Summer '25

INTERNSHIP

NASA Goddard Space Flight Center Intern |

Satellite-Based Mapping of Global Environmental Change

Summer 2020

- Programmed in JavaScript using the Google Earth Engine cloud-computing platform and API
- Collaborated with mentors to create four user-friendly web applications, analyzing satellite data (including the Landsat series) to visualize climate change impacts
- Identified and displayed changes in highly affected global locations
- Created an application to analyze precipitation trends by province in Ghana and Benin
- Designed an intuitive UI that allows users to observe ice melt, hurricane disturbances, and droughts

PROJECTS

MicroCaml (OCaml)

Fall 2024

- Developed a lexer, parser, optimizer, type-checker, and interpreter for a simplified programming language
- Regex powers lexing, a Context-Free Grammar structures parsing, Subtyping and Type-Checking rules verify well-typed expressions, and Operational Semantics define interpretation
- Optimizations include constant folding, constant propagation, and short-circuiting

A Shell (C)

Summer 2024

- Created a Unix-like shell that executes commands, manages processes, and supports input/output redirection
- Implemented control structures including pipes, subshells, and command conjunctions (AND, OR), enhancing command flexibility
- Managed file descriptors, enabling user-specified input and output files

Document Manager (C)

Summer 2024

- Implemented the back end of a simple text editor along with a text-based user interface
- Parsed user input strings to identify commands and execute them
- Developed complex methods using basic pointer operations, including one to identify and replace all instances of a target string

Other projects: Calendar application, NFA->DFA->Regex, personnel database modeling, a video rental manager, and more

Other experience: Lead guitarist for Lyn Lapid on two national tours (2022-2023), President of Animals for Animals at the University of Maryland (2022-2024), Co-Lead of March For Our Lives, Howard County (2019-2020)